

Selected Proofs in Analysis

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Metric spaces and complex variables

Based on individually authored Edinburgh assignments, February–March 2023. Revised 25 September 2026, with AI-assisted mathematical and editorial checking. These are expository proofs of established results; added arguments are identified within the text.

1 Metric spaces and complex analysis

1.1 Continuity characterized by closure

Source: Metric Spaces, Assignment 3, Question 1 (March 2023, p. 1).

This characterization expresses continuity through the way a map preserves limiting points. The distinction between an image and the closure of that image is essential.

Proposition 1.1. *Let $f : X \rightarrow Y$ be a map between metric spaces. Then f is continuous if and only if, for every subset $A \subseteq X$,*

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

The two closures are taken in X and Y , respectively.

Proof. Suppose first that f is continuous. Fix $A \subseteq X$ and put $C = \overline{f(A)}$. Since C is closed in Y , its inverse image $f^{-1}(C)$ is closed in X . Moreover, $A \subseteq f^{-1}(C)$. By the defining minimality of the closure,

$$\overline{A} \subseteq f^{-1}(C).$$

Applying f gives $f(\overline{A}) \subseteq C = \overline{f(A)}$.

Conversely, assume the stated inclusion holds for every $A \subseteq X$. Let $B \subseteq Y$ be closed and set $A = f^{-1}(B)$. We have $f(A) \subseteq B$, and hence

$$f(\overline{A}) \subseteq \overline{f(A)} \subseteq \overline{B} = B.$$

It follows that $\overline{A} \subseteq f^{-1}(B) = A$. Together with $A \subseteq \overline{A}$, this proves A is closed. Thus inverse images of closed sets are closed. Taking complements shows that inverse images of open sets are open, so f is continuous. $\square$

Why the inclusion matters. In general $f(f^{-1}(B)) = B \cap f(X)$, not B . For example, a constant map cannot have all of a two-point set as the image of its inverse image. Surjectivity is neither assumed nor needed in the proof.

1.2 Distance to a set and existence of a nearest point

Sources: Metric Spaces, Assignment 2, Question 2 (February 2023, pp. 1–2), and Assignment 3, Question 2 (March 2023, pp. 1–2). The Lipschitz formulation and the extension of the

nearest-point argument from $\mathbb{R}$ to arbitrary metric spaces are expository additions in this revision.

For a nonempty subset A of a metric space (X, d) , write

$$d_A(x) = \text{dist}(x, A) = \inf_{a \in A} d(x, a).$$

This infimum is a finite nonnegative real number: the distances are bounded below by zero and at least one finite distance is available.

Proposition 1.2. *The distance function d_A is 1-Lipschitz, and*

$$d_A = d_{\overline{A}}.$$

If $K \subseteq X$ is nonempty and compact, then for every $x \in X$ there is $y \in K$ such that $d(x, y) = d_K(x)$.

Proof. For $x, x' \in X$ and $a \in A$, the triangle inequality gives $d(x, a) \leq d(x, x') + d(x', a)$. Taking the infimum over $a \in A$ and then interchanging x and x' yields

$$|d_A(x) - d_A(x')| \leq d(x, x').$$

In particular, d_A is continuous.

Since $A \subseteq \overline{A}$, we have $d_{\overline{A}}(x) \leq d_A(x)$. For the reverse inequality, fix $y \in \overline{A}$. There is a sequence (a_n) in A with $d(a_n, y) \rightarrow 0$. Thus

$$d_A(x) \leq d(x, a_n) \leq d(x, y) + d(y, a_n).$$

Letting $n \rightarrow \infty$ gives $d_A(x) \leq d(x, y)$. Taking the infimum over $y \in \overline{A}$ proves the reverse inequality.

Now let K be compact and put $\delta = d_K(x)$. By the definition of infimum, choose $y_n \in K$ such that

$$\delta \leq d(x, y_n) < \delta + \frac{1}{n}.$$

Compactness in a metric space gives a subsequence $y_{n_j} \rightarrow y \in K$. The reverse triangle inequality implies $|d(x, y_{n_j}) - d(x, y)| \leq d(y_{n_j}, y) \rightarrow 0$. Hence $d(x, y) = \delta$. $\square$

Scope of the conclusion. Without compactness, attainment can fail: for $K = (0, 1) \subset \mathbb{R}$ and $x = 4$, the distance is 3, but no point of K attains it. This shows that one cannot simply omit the hypothesis; it does not assert that compactness is necessary in every example. Nor does compactness imply uniqueness: both points of $K = \{-1, 1\}$ are nearest to 0.

1.3 Holomorphic branches and a conformal map of a slit plane

Source: Complex Variables, Assignment 3, Questions 1–3 (February 2023, pp. 1–3).

Throughout this subsection, Log denotes the principal holomorphic logarithm on $\mathbb{C} \setminus (-\infty, 0]$, with imaginary part in $(-\pi, \pi)$. Choosing a branch means specifying a single-valued holomorphic function and checking its defining algebraic identity.

Proposition 1.3. *The function $z^2 - 1$ has holomorphic square roots on each of the domains $D = \{z : |z| < 1\}$ and $E = \{z : |z| > 1\}$. Explicit choices are*

$$\begin{aligned} s_D(z) &= i \exp\left(\frac{1}{2} \text{Log}(1 - z^2)\right), & z \in D, \\ s_E(z) &= z \exp\left(\frac{1}{2} \text{Log}(1 - z^{-2})\right), & z \in E. \end{aligned}$$

They satisfy $s_D(0) = i$ and $s_E(z)/z \rightarrow 1$ as $|z| \rightarrow \infty$. On each domain every holomorphic square root equals the displayed branch for that domain or its negative.

Proof. If $|w| < 1$, then $\operatorname{Re}(1 - w) \geq 1 - |w| > 0$. Taking $w = z^2$ on D and $w = z^{-2}$ on E shows that both logarithms are evaluated in the right half-plane, where the principal logarithm is holomorphic. The displayed functions are therefore holomorphic on their respective domains. Using $\exp(\operatorname{Log} u) = u$, we obtain

$$s_D(z)^2 = -(1 - z^2) = z^2 - 1, \quad s_E(z)^2 = z^2(1 - z^{-2}) = z^2 - 1.$$

The normalizations follow directly from continuity of the logarithm at 1.

Neither square root vanishes on its domain because ± 1 are excluded. If t is another holomorphic square root there, then t/s_D or t/s_E is a continuous function taking values only in $\{-1, 1\}$. Both D and E are connected, so this quotient is constant. This proves the final claim. $\square$

The exterior domain E is not simply connected. The explicit factorization above proves existence without incorrectly assuming that every nonvanishing holomorphic function on E has a logarithm.

Proposition 1.4. *Let $\Omega = \mathbb{C} \setminus (-\infty, -1]$. With the principal square root $\sqrt{z+1} = \exp(\frac{1}{2} \operatorname{Log}(z+1))$, the map*

$$F(z) = \frac{\sqrt{z+1} - 1}{\sqrt{z+1} + 1}$$

is a biholomorphic map from Ω onto the unit disc $\mathbb{D} = \{w : |w| < 1\}$. Its inverse is

$$F^{-1}(w) = \left(\frac{1+w}{1-w} \right)^2 - 1.$$

Proof. Translation by 1 maps Ω bijectively onto $\mathbb{C} \setminus (-\infty, 0]$. On this slit plane the principal square root maps bijectively onto the right half-plane $H = \{u : \operatorname{Re} u > 0\}$: its argument is half an angle in $(-\pi, \pi)$, and the inverse map on H is $u \mapsto u^2$.

For $u \in H$, the denominator $u + 1$ is nonzero and

$$|u + 1|^2 - |u - 1|^2 = 4 \operatorname{Re} u > 0.$$

Thus $M(u) = (u - 1)/(u + 1)$ belongs to $\mathbb{D}$. Conversely, for $w \in \mathbb{D}$, the denominator $1 - w$ is nonzero and

$$\operatorname{Re} \frac{1+w}{1-w} = \frac{1 - |w|^2}{|1-w|^2} > 0.$$

Consequently M maps H bijectively onto $\mathbb{D}$, with holomorphic inverse $(1+w)/(1-w)$. Composing these three maps and their inverses proves the result. In particular, if $u = \sqrt{z+1}$, then

$$F'(z) = \frac{1}{u(u+1)^2} \neq 0,$$

so the map is conformal at every point of Ω . $\square$

1.4 Cauchy estimates and growth rigidity of entire functions

Source: Complex Variables, Assignment 4, Solutions 1 and 3 (March 2023, pp. 1–2). The monomial conclusion below combines the general degree bound with the original quadratic example.

The growth condition in this exercise holds at every point of the plane, including near the origin. Its consequences therefore come from considering both large and small circles.

Proposition 1.5. *Let $N \geq 1$ be an integer and let f be entire. If there is $C > 0$ such that*

$$|f(z)| \leq C|z|^N \quad (z \in \mathbb{C}),$$

then $f(z) = az^N$ for some $a \in \mathbb{C}$ with $|a| \leq C$. In particular, $f^{(n)} \equiv 0$ for every $n > N$.

Proof. Write the Taylor expansion of f at 0 as $f(z) = \sum_{k=0}^{\infty} a_k z^k$. Since f is entire, this expansion is valid on all of $\mathbb{C}$. Cauchy's coefficient formula on a positively oriented circle of radius $R > 0$ gives

$$a_k = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta^{k+1}} d\zeta, \quad |a_k| \leq \frac{\max_{|\zeta|=R} |f(\zeta)|}{R^k} \leq CR^{N-k}.$$

If $k > N$, let $R \rightarrow \infty$ to obtain $a_k = 0$. If $k < N$, let $R \downarrow 0$ to obtain $a_k = 0$. For $k = N$, the same estimate gives $|a_N| \leq C$. Thus the only possibly nonzero Taylor coefficient is a_N , proving the claim. $\square$

The corresponding derivative estimate. At an arbitrary centre z_0 , Cauchy's formula reads

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|\zeta-z_0|=R} \frac{f(\zeta)}{(\zeta-z_0)^{n+1}} d\zeta.$$

It gives the valid bound

$$|f^{(n)}(z_0)| \leq n!C \frac{(|z_0| + R)^N}{R^n} = n!CR^{N-n} \left(1 + \frac{|z_0|}{R}\right)^N.$$

For $n > N$, the right-hand side tends to zero as $R \rightarrow \infty$. The factor $(1 + |z_0|/R)^N$ tends to 1; it must not be discarded as though it were at most 1.

If a polynomial growth bound is assumed only outside a disc, it still forces f to be a polynomial of degree at most N , but the small-circle argument is unavailable. The stronger monomial conclusion uses the global bound in the proposition.

1.5 Uniform convergence and holomorphy of a Dirichlet series

Source: Complex Variables, Assignment 4, Solution 4 (March 2023, p. 3). The explicit remainder estimate is an addition in this revision.

For an integer $n \geq 1$, define $n^{-z} = \exp(-z \log n)$ using the real logarithm of the positive number n . This removes any ambiguity about branches.

Proposition 1.6. *The series*

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$

converges absolutely and locally uniformly on $\{z \in \mathbb{C} : \operatorname{Re} z > 1\}$, and defines a holomorphic function there. More precisely, for each $\varepsilon > 0$, it converges uniformly on $U_\varepsilon = \{z : \operatorname{Re} z > 1 + \varepsilon\}$, with

$$\sup_{z \in U_\varepsilon} \left| \zeta(z) - \sum_{n=1}^M n^{-z} \right| \leq \frac{M^{-\varepsilon}}{\varepsilon} \quad (M \geq 1).$$

Proof. Write $z = x + iy$. Because $|e^{-iy \log n}| = 1$, we have

$$|n^{-z}| = e^{-x \log n} = n^{-x}.$$

For $z \in U_\varepsilon$ this is at most $n^{-(1+\varepsilon)}$. The latter terms form a convergent positive series, so the Weierstrass M -test gives absolute and uniform convergence on U_ε . The integral comparison for the decreasing function $t \mapsto t^{-(1+\varepsilon)}$ gives

$$\sum_{n=M+1}^{\infty} n^{-(1+\varepsilon)} \leq \int_M^{\infty} t^{-(1+\varepsilon)} dt = \frac{M^{-\varepsilon}}{\varepsilon},$$

which proves the stated remainder bound.

Each summand $z \mapsto \exp(-z \log n)$ is entire. Given any z_0 with $\operatorname{Re} z_0 > 1$, set $\varepsilon = (\operatorname{Re} z_0 - 1)/2 > 0$. Then U_ε is an open neighbourhood of z_0 , and the holomorphic partial sums converge uniformly there. The theorem that a locally uniform limit of holomorphic functions is holomorphic now shows that ζ is holomorphic near z_0 . As z_0 was arbitrary, the conclusion holds throughout the half-plane. $\square$

This argument establishes the series only in its half-plane of absolute convergence. It makes no claim about analytic continuation or zeros outside that half-plane.